

ST GEORGE ISLAND, AK, USA

WMO: 703835

Lat: 56.600N

Lon: 169.565W

Elev: 38

StdP: 100.87

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 25628

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-13.9	-12.3	-17.2	0.8	-13.3	-15.7	1.0	-11.7	21.1	-4.4	19.7	-2.0	8.0	340	0.990

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.9	13.0	11.4	12.3	11.0	11.5	10.5	11.9	12.5	11.4	11.9	10.8	11.2	6.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	8.5	12.2	11.1	8.2	11.7	10.3	7.8	11.0	34.2	12.5	32.6	11.8	31.4	11.3	16.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.5	15.1	DB	-14.6	15.5	3.5	2.5	-17.2	17.3	-19.2	18.8	-21.2	20.2	-23.8	22.0	(4)
(5)			WB	-14.6	12.8	3.2	1.5	-17.0	13.9	-18.9	14.8	-20.7	15.7	-23.0	16.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-2.4	-1.6	-2.4	-0.1	2.8	6.1	8.7	9.7	7.9	4.4	1.2	-1.0
	DBStd	5.25	4.18	4.41	4.36	3.16	2.42	1.70	1.72	1.50	1.88	2.39	2.67	3.57
	HDD10.0	2652	385	325	386	302	223	116	47	24	66	174	263	341
	HDD18.3	5668	643	558	644	552	482	366	298	268	313	433	513	599
	CDD10.0	23	0	0	0	0	0	0	7	14	2	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.4	9.4	9.1	8.4	7.8	6.7	5.7	5.0	5.4	6.7	7.8	8.7	8.6
Precipitation	PrecAvg	1060	96	95	75	52	59	69	99	96	109	90	111	109
	PrecMax	1244	163	190	163	94	101	107	149	135	182	154	177	173
	PrecMin	833	41	32	25	24	26	22	41	38	58	48	49	61
	PrecStd	100	29	34	33	16	19	24	27	20	29	23	33	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	5.1	4.7	6.2	9.6	12.6	15.2	14.6	12.8	9.7	7.4	6.1
		MCWB	3.9	4.6	3.7	4.4	6.8	9.5	11.8	12.5	11.6	9.0	6.4	5.2
	2%	DB	3.8	4.1	3.9	5.2	7.8	10.5	12.9	13.1	11.8	9.0	6.5	4.9
		MCWB	3.1	3.4	3.1	4.1	5.9	8.5	11.1	11.7	10.6	8.3	5.6	4.2
	5%	DB	3.0	3.5	3.0	4.3	6.9	9.4	12.0	12.4	11.1	8.4	6.0	4.0
		MCWB	2.3	2.8	2.2	3.4	5.4	8.0	10.7	11.4	10.1	7.6	5.2	3.3
	10%	DB	2.5	2.9	2.4	3.7	6.1	8.5	11.1	11.9	10.3	7.6	5.1	3.3
		MCWB	1.9	2.2	1.7	2.9	4.9	7.4	10.2	11.1	9.4	6.7	4.4	2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.9	4.4	5.2	7.5	10.1	12.7	13.1	12.0	9.5	7.0	5.6
		MCDB	4.4	5.0	4.8	5.6	9.0	11.7	14.1	14.0	12.3	9.6	7.1	5.9
	2%	WB	3.3	3.7	3.4	4.3	6.4	9.0	11.7	12.3	11.0	8.7	6.2	4.5
		MCDB	3.6	3.9	3.7	4.9	7.3	10.1	12.5	12.8	11.3	8.9	6.5	4.8
	5%	WB	2.7	3.0	2.6	3.7	5.7	8.1	10.9	11.7	10.3	7.9	5.3	3.6
		MCDB	3.0	3.3	3.0	4.3	6.6	9.0	11.6	12.1	10.7	8.2	5.8	3.9
	10%	WB	2.0	2.4	1.8	3.0	5.1	7.7	10.3	11.2	9.8	7.2	4.4	2.8
		MCDB	2.3	2.8	2.3	3.6	6.0	8.4	10.9	11.6	10.2	7.6	5.1	3.2
Mean Daily Temperature Range	5% DB	MDBR	3.4	3.7	3.7	3.6	3.7	3.6	3.1	2.9	3.2	3.4	3.3	3.8
		MCDBR	2.6	3.0	3.6	3.7	4.9	5.1	4.6	3.8	3.6	2.9	3.5	3.6
		MCWBR	2.7	3.3	3.3	3.2	3.6	3.5	3.3	3.1	3.0	2.8	3.7	3.6
	5% WB	MCDBR	2.7	3.2	3.4	3.5	4.4	4.5	4.0	3.3	2.9	2.5	3.4	3.3
		MCWBR	2.9	3.6	3.4	3.1	3.4	3.3	3.1	3.0	2.6	2.8	3.7	3.6
Clear-Sky Solar Irradiance	taub	0.302	0.340	0.335	0.399	0.405	0.389	0.395	0.388	0.370	0.340	0.324	0.274	
	taud	2.226	2.234	2.333	2.138	2.217	2.366	2.390	2.412	2.365	2.317	2.167	2.340	
	Ebn at Noon	575	699	824	809	832	853	839	820	776	694	521	539	
	Edh at Noon	54	80	93	131	129	113	108	100	90	73	56	38	
All-Sky Solar Radiation	RadAvg	0.42	1.01	2.29	3.53	4.17	4.23	3.65	3.09	2.42	1.45	0.56	0.29	
	RadStd	0.05	0.07	0.36	0.37	0.31	0.28	0.29	0.28	0.17	0.08	0.05	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page